

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development Practical
Project

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Class:TYIT	Batch:2
Date of Assignment: 23/09/26	Date/Time of Submission: 30/09/26

AI-Powered Heart Disease Risk Analysis and Prediction

Description:

AI-Based Heart Disease Risk Prediction System Using Machine Learning is an intelligent application designed to predict the risk of heart disease based on health-related factors such as age, blood pressure, cholesterol level, BMI, smoking habits, and physical activity. The system uses a Random Forest classification algorithm to analyze the given inputs and predict the risk level. It also displays the risk probability, accuracy, precision, recall, F1-score, and important contributing factors, making the system useful for understanding how machine learning can be applied to healthcare risk prediction.

Problem Statement / Motivation:

Heart disease is a major health concern, and identifying risk factors at an early stage can be helpful. Traditional assessment may require analyzing multiple health-related factors such as age, blood pressure, cholesterol, BMI, smoking habits, and physical activity. Therefore, there is a need for an easy-to-use system that can analyze these factors and provide an estimated risk level. This project uses Artificial Intelligence and Machine Learning to develop a system that predicts heart disease risk from user-provided health information. The main motivation is to demonstrate how machine learning can support early risk assessment and help users understand important factors associated with the prediction.

Approach and Technologies Used:

Approach:

1. Collect health-related input such as age, blood pressure, cholesterol, BMI, smoking, and physical activity.
2. Prepare the data and divide it into training and testing datasets.
3. Train a Random Forest Classification model using the training data.
4. Use the trained model to predict the user's heart disease risk.
5. Display the risk probability, risk level, and important factors.
6. Evaluate the model using Accuracy, Precision, Recall, and F1 Score.

Technologies Used:

- Python – Programming language
- Pandas – Data handling and processing
- Scikit-learn – Machine learning model and evaluation
- Random Forest – Classification algorithm
- Streamlit – Web-based user interface
- VS Code – Development environment

Code Snippet:

```
# Train Random Forest Model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
model.fit(X_train, y_train)
# Predict heart disease risk
prediction = model.predict(input_data)[0]
# Calculate risk probability
probability = model.predict_proba(input_data)[0][1]
# Display result
if prediction == 1:
    st.error(" ⚠ Higher Heart Disease Risk Detected")
else:
    st.success(" ✅ Lower Heart Disease Risk Detected")
st.write(f'Estimated Risk Probability: {probability * 100:.2f}%')
```

Output Screenshot:-

```
● PS C:\Users\mvluc\Desktop\AI_Heart_Risk_Prediction> pip install streamlit pandas scikit-learn
Requirement already satisfied: streamlit in C:\Users\mvluc\AppData\Local\Programs\Python\Python314\Lib\site-packages (1.64.0)
Requirement already satisfied: pandas in C:\Users\mvluc\AppData\Local\Programs\Python\Python314\Lib\site-packages (3.0.5)
Requirement already satisfied: scikit-learn in C:\Users\mvluc\AppData\Local\Programs\Python\Python314\Lib\site-packages (1.9.0)
Requirement already satisfied: altair!=5.4.0,!=5.4.1,<7,>=5.0.0 in C:\Users\mvluc\AppData\Local\Programs\Python\Python314\Lib\site-packages (6.3.0)
PS C:\Users\mvluc\Desktop\AI_Heart_Risk_Prediction> streamlit run app.py

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2026-09-30 12:15:51.958 Uvicorn server started on :::8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.29.125:8501
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For `use_container_width=True`, use `width='stretch'`. For `use_container_width=False`, use `width='content'`.
```

localhost:8501
Deploy

AI-Based Heart Disease Risk Prediction

An educational AI system that predicts heart disease risk using a Random Forest classification model. A Group Project Made By Raj, Khushi, Priti

Enter Patient Information

Age
40

BMI
24.00

Blood Pressure
120

Smoking
No

Cholesterol
200

Regular Physical Activity
Yes

Predict Heart Disease Risk

AI Model Performance

Accuracy
100.00%

Precision
100.00%

Recall
100.00%

F1 Score
100.00%

Training Dataset

View Training Dataset

About This Project

This project uses a Random Forest Classification algorithm to predict heart disease risk based on age, blood pressure, cholesterol, BMI, smoking habits and physical activity.

The system also displays risk probability, risk level, feature importance and model performance metrics.

This project is created for educational purposes only. It is not a medical diagnosis and should not replace professional medical advice.

localhost:8501
Deploy

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GitHub Repository:

<https://github.com/Rajit0311/AI-Heart-Risk-Predication.git>

Challenges Faced:

- Finding and preparing suitable health-related data for training the model.
- Handling and converting user inputs into the correct format required by the machine learning model.
- Training the Random Forest model and achieving reliable prediction results.
- Integrating the machine learning model with the Streamlit interface.
- Displaying the risk probability, risk level, and model performance in a clear and understandable way.
- Ensuring the system is used only for educational risk prediction and not medical diagnosis.

Future Improvements:

- Add support for multiple patients instead of predicting for only one patient at a time.
- Create a Graphical User Interface (GUI) for a better user experience.
- Store patient data and prediction results in a file or database.
- Add input validation to avoid incorrect or invalid health information.
- Include more health features such as family history, diabetes, exercise habits, and other relevant factors.
- Add a login system to improve data security and privacy.
- Use a larger and real-world dataset to improve the reliability of the machine learning model.

Learning Outcome:

- Learned how to apply Artificial Intelligence and Machine Learning to a real-world healthcare problem.
- Gained practical knowledge of the Random Forest Classification algorithm.
- Learned how to prepare data, train a model, and make predictions.
- Understood how to evaluate a machine learning model using Accuracy, Precision, Recall, and F1 Score.
- Learned to develop an interactive web application using Python and Streamlit.
- Understood the importance of data validation, privacy, and responsible use of AI in healthcare applications.

Conclusion:

The AI-Based Heart Disease Risk Prediction System successfully demonstrates the use of Artificial Intelligence and Machine Learning in healthcare risk assessment. The Random Forest algorithm analyzes health-related factors and provides an estimated risk level along with probability and important contributing factors. The project also provides practical experience in data processing, machine learning, model evaluation, and Streamlit application development. Overall, it shows how AI can be applied to develop an interactive system for educational health-risk analysis.